

POWERSHELL TERMINAL

Here is where your presentation begins

TABLE OF CONTENTS

01 Command Structure

02 Basic Commands

03 Variables

04 Piping

01

COMMAND STRUCTURE

INTRODUCTION

English is one of the most widely spoken languages, falling third to only Mandarin and Spanish. English is widely used throughout a multitude of programming languages to describe key actions or descriptors.





VERB-NOUN

POWERSHELL STRUCTURE



THE VERB–NOUN STRUCTURE

Powershell uses Verb-Noun structure. A verb describes what it does, a noun describes what it is. Get-EventLog retrieves the event log.

Java Syntax = verbNoun = Troy.getName()
Powershell Syntax = Verb-Noun = Get-Name Troy

COMMON VERBS

VERBS

A word that describes action or movement



NEW
Creates a new NOUN

GET
Retrieves an existing NOUN

SET
Edits a NOUN object

START
Starts up a new NOUN

02

BASIC COMMANDS

VOCABULARY

CMDLET

Commands built into shell written in .NET

PARAMETER

Argument to a Cmdlet/Function/Script

FUNCTION

Commands written in PowerShell language

ALIAS

Shortcut for a Cmdlet or Function

SCRIPT

Text files with .ps1 extension

APPLICATION

Existing windows programs

COMMON CMDLETS

CMD Commands - Same function, different interpretation as aliases

cd/chdir/sl = Set-Location, Set the current working location
pwd/gl = Get-Location, Get and display the current location
dir/ls/gci = Get-ChildItem, Get child items (contents of a folder or registry key)
cls/clear = Clear-Host, Clear the screen
cat/gc/type = Get-Content, Get content from item (specific location)
ac = Add-Content, Add to the content of the item
sc = Set-Content, Set content in the item (specific location)
copy/cp/cpi = Copy-Item, Copy an item from a namespace location
del/erase/rd/ri/rm/rmdir = Remove-Item, Remove an item
mi/move/mv = Move-Item, Move an item from one location to another
si = Set-Item, Change the value of an item
md/mkdir/ni = New-Item, Create a new item in a namespace
sleep = Start-Sleep, Suspend shell, script, or runspace activity
compare/diff = Compare-Diff, Compare the properties of objects

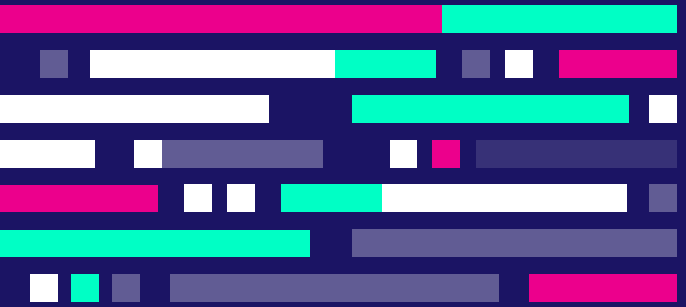
COMMON CMDLETS

CMD Commands - Same function, different interpretation as aliases

group = Group-Object, Group objects that contain the same value
curl/iwr/wget = Invoke-WebRequest, Get content from a web page
measure = Measure-Object, Measure the properties of an object
nal = New-Alias, Create a new alias
set/sv = Set-Variable, Set a variable and a value
shcm = Show-Command, Create PowerShell commands in a graphical command window
sort = Sort-Object, Sort objects by property value
sasv = Start-Service, Start a stopped service
saps/start = Start-Process, Start one or more processes
spsv = Stop-Service, Stop a running service
kill/spps = Stop-Process, Stop a running process
?/where = Where-Object, Filter the objects passed along the command pipeline
echo/write = Write-Output, Write an object to the pipeline

03

VARIABLES



VARIABLE INTRODUCTION

All variables in Powershell are objects. They act similar to python variables.

Can be set innately or through commands.
$A = “Alphabet” creates a new variable “A”
New-Variable A -Value “Alphabet” also creates a new variable “A”

Variable Tips

$a, $b = 0 or **$a,$b = ‘a’,’b’** - Assigns multiple variables
$a, $b = $b, $a - Flips Variables
$var = [int]5 - Signs 5 as an integer
$var += 1 - Adds 1 to var
$var -= 1 - Subtracts 1 to var
$arr = “a”,”b” - Array of strings
$arr = @() - Empty array
$arr[5] - 5th array element
$arr[-3..-1] - Last 3 array elements

04

PIPELINE

PIPELINE SYMBOL:

How does it work?

COMMAND ONE

The first command runs and returns its output

COMMAND TWO

The output from Command One is treated as input for Two

COMMON USAGES

Get-ChildItem | Select-String “Programs”

- **Returns only files that say Program**

Get-Variable path | Set-Location

- **Navigates to the location at $path**

Invoke-WebRequest “google.com” | Out-File “google.html”

- **Writes to an HTML file based on google.com**

Get-EventLog -List | Where-Object { $_.Log -eq "Application" }

- **Gets all logs with the Application tag**

THANKS !



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